Frobenius and the Point Counts of an Abelian Variety
Chapter II, Theorem 1.1

J.S. Milne

Private reading transcription of AV v2.00 (2008)
with separate supporting material from the corrected Storrs article, January 2, 2022

Fixed sources and selection. The primary source is J.S. Milne, *Abelian Varieties*, course notes, 172 physical pages. Its cover reads Version 2.0, March 16, 2008; its bibliographic version is 2.00 and its internal history also records March 17, 2008. These are the printed labels of the fixed copy, not an inferred replacement version. The source URL is `https://www.jmilne.org/math/CourseNotes/AV.pdf`, with landing page `https://www.jmilne.org/math/CourseNotes/av.html`. The primary selection is Chapter II, Section 1, physical pages 81–84 (printed pages 75–78), through the complete proofs of Lemmas 1.2 and 1.3. Remark 1.4 is retained as uncounted closing context. The zeta-function corollary following it and the Honda–Tate section are outside this selection.

The separate supporting source is Milne's corrected author file for *Abelian Varieties*, originally Chapter V of *Arithmetic Geometry* (Storrs, Conn., 1984), pp. 103–150, Springer, New York, 1986. The actual fixed corrected PDF is dated **January 2, 2022** and has 49 physical pages. Its URL is `https://www.jmilne.org/math/xnotes/AVs.pdf`, with author page `https://www.jmilne.org/math/xnotes/AVs.html`. Only Section 17's necessary context, Theorem 17.3, Lemma 17.4 and their complete printed argument on physical/printed pages 35–36 are appended. The 1986 publication, this corrected 2022 file, and the 2008 course notes remain separately identified sources.

Editorial note. There is one primary target, II Theorem 1.1, for arbitrary abelian varieties over a finite field, including non-simple ones, and all integers $m \geq 1$. No restriction excluding characteristics 2 or 3 is added. II Lemmas 1.2 and 1.3 are retained as substantive steps of the author's primary proof. I Proposition 10.20, I Corollary 10.24 and Rosati positivity are explicit strong supporting inputs. Supporting statements, definitions and the separately sourced Rosati calculation are not additional primary targets. No new proof is supplied.

Necessary notation and characteristic-polynomial inputs from the 2008 notes

Source: I, Section 10, physical page 49, printed page 43; notation excerpt.

We let $\mathrm{End}^0(A) = \mathrm{End}(A) \otimes \mathbb{Q}$.

Source: I, Section 10, physical page 50, printed page 44; opening paragraph.

Let A be an abelian variety of dimension g over a field k. Recall that, for any m not divisible by the characteristic of k, $A_m(k^{\mathrm{sep}})$ has order m^{2g}, and that, for a prime $\ell \neq \mathrm{char}(k)$, $T_\ell A = \varprojlim A_{\ell^n}(k^{\mathrm{sep}})$ (see 7.3).

Source: I Proposition 10.5 and following paragraph, physical page 51, printed page 45.

Proposition 10.5. *For $\ell \neq \mathrm{char}(k)$, $T_\ell A$ is a free $\mathbb{Z}_\ell$-module of rank $2g$.*

A homomorphism $\alpha : A \to B$ induces a homomorphism $A_n(k^{\text{sep}}) \to B_n(k^{\text{sep}})$, and hence a homomorphism

$$T_\ell\alpha : T_\ell A \longrightarrow T_\ell B, \qquad (a_1, a_2, \ldots) \longmapsto (\alpha(a_1), \alpha(a_2), \ldots).$$

Therefore T_ℓ is a functor from abelian varieties to $\mathbb{Z}_\ell$-modules.

Source: I, Section 10, physical page 52, printed page 46; rational Tate-space definition.

Write $V_\ell A = T_\ell A \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ $(= T_\ell A \otimes_{\mathbb{Z}} \mathbb{Q})$.

Source: I Theorem 10.9 and its immediately preceding degree convention, physical page 52, printed page 46.

Let $\alpha \in \text{End}(A)$. If α is an isogeny, we define $\deg(\alpha)$ as before; otherwise, we set $\deg(\alpha) = 0$.

Theorem 10.9. *Let $\alpha \in \text{End}(A)$. There is a unique monic polynomial $P_\alpha \in \mathbb{Z}[X]$ of degree $2g$ such that $P_\alpha(r) = \deg(\alpha - r)$ for all integers r.*

Source: I Remark 10.11, physical page 53, printed page 47; degree and polynomial extension.

For $\alpha \in \text{End}(A)$ and $n \in \mathbb{Z}$,

$$\deg(n\alpha) = \deg(n_A) \cdot \deg(\alpha) = n^{2g} \deg(\alpha).$$

We can use this formula to extend the definition of deg to $\text{End}^0(A)$. Since $\text{End}(A)$ is torsion-free, we can identify $\text{End}(A)$ with a submodule of $\text{End}^0(A)$. For $\alpha \in \text{End}^0(A)$, define

$$\deg(\alpha) = n^{-2g} \deg(n\alpha)$$

if n is any integer such that $n\alpha \in \text{End}(A)$. The previous formula shows that this is independent of the choice of n. Similarly, once we have proved the theorem, we can define

$$P_\alpha(X) = n^{-2g} P_{n\alpha}(nX), \qquad \alpha \in \text{End}^0(A), \quad n\alpha \in \text{End}(A).$$

Then $P_\alpha(X)$ is a monic polynomial of degree $2g$ with rational coefficients, and

$$P_\alpha(r) = \deg(\alpha - r), \qquad \text{any } r \in \mathbb{Q}.$$

Source: I, Section 10, physical page 54, printed page 48; trace convention.

We call P_α the *characteristic polynomial* of α and we define the *trace* $\text{Tr}(\alpha)$ of α by the equation

$$P_\alpha(X) = X^{2g} - \text{Tr}(\alpha) X^{2g-1} + \cdots + \deg(\alpha).$$

Source: I Proposition 10.20, physical page 56, printed page 50.

Proposition 10.20. *For all $\ell \neq \text{char}(k)$, $P_\alpha(X)$ is the characteristic polynomial of α acting on $V_\ell A$; hence the trace and degree of α are the trace and determinant of α acting on $V_\ell A$.*

Source: I Corollary 10.24, physical page 59, printed page 53.

Corollary 10.24. *Let $\alpha \in \text{End}^0(A)$, and assume $\mathbb{Q}[\alpha]$ is a product of fields. Let $C_\alpha(X)$ be the characteristic polynomial of α acting on $\mathbb{Q}[\alpha]$ (e.g., if $\mathbb{Q}[\alpha]$ is a field, this is the minimum polynomial of α); then*

$$\{a \in \mathbb{C} \mid C_\alpha(a) = 0\} = \{a \in \mathbb{C} \mid P_\alpha(a) = 0\}.$$

Polarization and the parent Rosati input

Source: I, Section 11, physical page 59, printed page 53; polarization definition.

A *polarization*[15] λ of an abelian variety A is an isogeny $\lambda : A \to A^\vee$ such that, over k^{al}, λ becomes of the form $\lambda_{\mathcal{L}}$ for some ample sheaf $\mathcal{L}$ on $A_{k^{\mathrm{al}}}$. Unfortunately, this is not quite the same as requiring that λ itself be of the form $\lambda_{\mathcal{L}}$ for $\mathcal{L}$ an ample invertible sheaf on A (Milne 1986, 13.2).

Source: I, Section 14, physical page 67, printed page 61; Rosati definition.

Fix a polarization λ on A. As λ is an isogeny $A \to A^\vee$, it has an inverse in $\mathrm{Hom}^0(A^\vee, A) \stackrel{\mathrm{df}}{=} \mathrm{Hom}(A^\vee, A) \otimes \mathbb{Q}$. The *Rosati involution* on $\mathrm{End}(A) \otimes \mathbb{Q}$ corresponding to λ is

$$\alpha \longmapsto \alpha^\dagger = \lambda^{-1} \circ \alpha^\vee \circ \lambda.$$

Source: I Theorem 14.3 and its complete printed proof, physical page 68, printed page 62.

The next result is very important.

Theorem 14.3. *The bilinear form*

$$(\alpha, \beta) \longmapsto \mathrm{Tr}(\alpha \circ \beta^\dagger) : \mathrm{End}(A) \otimes \mathbb{Q} \times \mathrm{End}(A) \otimes \mathbb{Q} \longrightarrow \mathbb{Q}$$

is positive definite, i.e., $\mathrm{Tr}(\alpha \circ \alpha^\dagger) > 0$ *for* $\alpha \neq 0$. *More precisely, let* D *be the ample divisor defining the polarization used in the definition of* $\dagger$*; then*

$$\mathrm{Tr}(\alpha \circ \alpha^\dagger) = \frac{2g}{(D^g)}(D^{g-1} \cdot \alpha^*(D)).$$

Proof. As D is ample and $\alpha^*(D)$ is effective, the intersection number $(D^{g-1} \cdot \alpha^*(D))$ is positive. Thus the second statement implies the first. The second statement is proved by a calculation, which we omit for the present (see Milne 1986, §17). □

Editorial note. The preceding omission is part of the 2008 source and is retained literally. The separate 2022 author calculation appended below supports precisely this trace/intersection identity; it is not represented as a calculation printed in the 2008 parent. The unrelated I13.12 polarization issue is not part of this selection. The statements of I10.20 and I10.24 are retained as explicit inputs; their full preceding chapter is not transcribed.

Chapter II. Abelian Varieties: Arithmetic

1. The Zeta Function of an Abelian Variety

Source: 2008 notes, physical page 81, printed page 75; complete introduction to II1.1.

We write $\mathbb{F}_q$ for a finite field with q elements, $\mathbb{F}$ for an algebraic closure of $\mathbb{F}_q$, and $\mathbb{F}_{q^m}$ for the unique subfield of $\mathbb{F}$ with q^m elements. Thus the elements of $\mathbb{F}_{q^m}$ are the solutions of $c^{q^m} = c$.

For a variety V over $\mathbb{F}_q$, the Frobenius map $\pi_V : V \to V$ is defined to be the identity map on the underlying topological space of V and is the map $f \mapsto f^q$ on $\mathcal{O}_V$. For example, if $V = \mathbb{P}^n = \mathrm{Proj}(k[X_0, \ldots, X_n])$, then π_V is defined by the homomorphism of rings

$$X_i \longmapsto X_i^q : k[X_0, \ldots, X_n] \longrightarrow k[X_0, \ldots, X_n]$$

[15] This notion of polarization differs slightly from Weil's original definition.

and induces the map on points

$$(x_0 : \cdots : x_n) \longmapsto (x_0^q : \cdots : x_n^q) : \mathbb{P}^n(\mathbb{F}) \longrightarrow \mathbb{P}^n(\mathbb{F}).$$

For any regular map $\varphi : W \to V$ of varieties over $\mathbb{F}_q$, it is obvious that $\varphi \circ \pi_W = \pi_V \circ \varphi$. Therefore, if $V \hookrightarrow \mathbb{P}^n$ is a projective embedding of V, then π_V induces the map

$$(x_0 : \ldots : x_n) \longmapsto (x_0^q : \ldots : x_n^q)$$

on $V(\mathbb{F})$. Thus $V(\mathbb{F}_q)$ is the set of fixed points of $\pi_V : V(\mathbb{F}) \to V(\mathbb{F})$.

Let A be an abelian variety over $\mathbb{F}_q$. Then π_A maps 0 to 0 (because $0 \in V(\mathbb{F}_q)$), and so it is an endomorphism of A. Recall that its characteristic polynomial P_π is a monic polynomial of degree $2g$, $g = \dim(A)$, with coefficients in $\mathbb{Z}$.

Editorial note. In the error-bound proof on parent physical page 82, the source prints $(2^g - 2g - 1)$ for the remaining terms; the theorem statement on physical page 81 uses 2^{2g}. Both source expressions are retained in their own locations. This note records their printed difference and does not assert an author-issued correction.

Source: 2008 notes, II Theorem 1.1, physical page 81, printed page 75.

Theorem 1.1. *Write $P_\pi(X) = \prod_i (X - a_i)$, and let $N_m = |A(\mathbb{F}_{q^m})|$. Then*

(a) $N_m = \prod_{i=1}^{2g} (1 - a_i^m)$ for all $m \geq 1$, and

(b) (Riemann hypothesis) $|a_i| = q^{1/2}$.

Hence

$$|N_m - q^{mg}| \leq 2g \cdot q^{m(g-\frac{1}{2})} + (2^{2g} - 2g - 1) q^{m(g-1)}.$$

Source: 2008 notes, physical pages 81–82, printed pages 75–76; complete initial proof of II1.1.

Proof. We first deduce the inequality from the preceding statements. Take $m = 1$ in (a) and expand out to get

$$|A(\mathbb{F}_q)| = a_1 \cdots a_{2g} - \sum_{i=1}^{2g} a_1 \cdots a_{i-1} a_{i+1} \cdots a_{2g} + \cdots .$$

The first term on the right is an integer, and in fact a positive integer because it is $P_\pi(0) = \deg(\pi)$, and (b) shows that it has absolute value q^g. Hence it equals q^g (actually, it easy to prove directly that $\deg(\pi) = q^g$). The Riemann hypothesis shows that each term $a_1 \cdots a_{i-1} a_{i+1} \cdots a_{2g}$ has absolute value $= q^{g-\frac{1}{2}}$, and so the sum has absolute value $\leq 2g \cdot q^{g-\frac{1}{2}}$. There are $(2^g - 2g - 1)$ terms remaining, and each has absolute value $\leq q^{g-1}$, whence the inequality. We first prove (a) in the case $m = 1$. The kernel of

$$\pi - \mathrm{id} : A(\mathbb{F}) \longrightarrow A(\mathbb{F})$$

is $A(\mathbb{F}_q)$. I claim that the map

$$(d\pi)_0 : \mathrm{Tgt}_0(A) \longrightarrow \mathrm{Tgt}_0(A)$$

is zero—in fact, that this is true for any variety. In proving it, we can replace A with an open affine neighbourhood U, and embed U into $\mathbb{A}^m$ some m in such a way that 0 maps to the origin 0. The

map $(d\pi)_0$ on $\mathrm{Tgt}_0(U)$ is the restriction of the map $(d\pi)_0$ on $\mathrm{Tgt}_0(\mathbb{A}^m)$. But $\pi : \mathbb{A}^n \to \mathbb{A}^n$ is given by the equations $Y_i = X_i^q$, and $d(X_i^q) = qX_i^{q-1} = 0$ (in characteristic p). We now find that

$$d(\pi - \mathrm{id})_0 = (d\pi)_0 - (d(\mathrm{id}))_0 = -1.$$

Hence $\pi - 1$ is *étale* at the origin, and so, by homogeneity, it is *étale* at every point—each point in the kernel occurs with multiplicity 1. Therefore,

$$|A(\mathbb{F}_q)| = \deg(\pi - \mathrm{id}).$$

But, from the definition of P_π, we know that

$$\deg(\pi - \mathrm{id}) = P_\pi(1),$$

and this is $\prod(1 - a_i)$. When we replace π with π^m in the above argument, we find that

$$|A(\mathbb{F}_{q^m})| = P_{\pi^m}(1).$$

Recall (10.20) that $a_1, \ldots, a_{2g}$ can be interpreted as the eigenvalues of π acting on $T_\ell A$. Clearly π^m has eigenvalues $a_1^m, \ldots, a_{2g}^m$, and so

$$P_{\pi^m}(X) = \prod(X - a_i^m), \qquad P_{\pi^m}(1) = \prod(1 - a_i^m)$$

which proves (a) for a general m. □

Part (b) follows from the next two lemmas.

Source: 2008 notes, II Lemma 1.2 and complete proof, physical pages 82–83, printed pages 76–77.

Lemma 1.2. *Let † be the Rosati involution on* $\mathrm{End}(A) \otimes \mathbb{Q}$ *defined by a polarization of A; then* $\pi_A^\dagger \circ \pi_A = q_A$.

Proof. Let D be the ample divisor on A defining the polarization; thus $\lambda(a) = [D_a - D]$. We have to show that

$$\pi^\vee \circ \lambda \circ \pi = q\lambda.$$

Recall that, on points, $\pi^\vee$ is the map

$$[D'] \longmapsto [\pi^* D'] : \mathrm{Pic}^0(A) \longrightarrow \mathrm{Pic}^0(A).$$

Let D' be a divisor on A (or, in fact any variety defined over $\mathbb{F}_q$). If $D' = \mathrm{div}(f)$ near $\pi(P)$, then, by definition, $\pi^* D' = \mathrm{div}(f \circ \pi)$ near P. But $\pi(P) = P$ and $f \circ \pi = f^q$ (this was the definition of π), and $\mathrm{div}(f^q) = q\,\mathrm{div}(f)$; thus $\pi^* D' = qD$. Next observe that, for any homomorphism $\alpha : A \to A$ and any point a on A,

$$\alpha \circ t_a(x) = \alpha(a + x) = \alpha(a) + \alpha(x) = t_{\alpha(a)} \circ \alpha(x).$$

We can now prove the lemma. For any $a \in A(\mathbb{F})$, we have

$$\begin{aligned}
(\pi^\vee \circ \lambda \circ \pi)(a) &= \pi^*[D_{\pi(a)} - D] \\
&= [\pi^* t_{\pi(a)}^* D - \pi^* D] \\
&= [(t_{\pi(a)} \circ \pi)^* D - \pi^* D] \\
&= [(\pi \circ t_a)^* D - \pi^* D] \\
&= [t_a^* \pi^* D - \pi^* D] \\
&= [t_a^* qD - qD] \\
&= q\lambda(a);
\end{aligned}$$

as required. □

Source: 2008 notes, II Lemma 1.3 and complete proof, physical pages 83–84, printed pages 77–78, including original footnote 1.

Lemma 1.3. *Let A be an abelian variety over a field k (not necessarily finite). Let α be an element of* $\mathrm{End}(A) \otimes \mathbb{Q}$ *such that $\alpha^\dagger \circ \alpha$ is an integer r; for any root a of P_α in $\mathbb{C}$, $|a|^2 = r$.*

Proof. Note that $\mathbb{Q}[\alpha]$ is a commutative ring of finite-dimension over $\mathbb{Q}$; it is therefore an Artin ring. According to (Atiyah and Macdonald 1969, Chapter 8)[1], it has only finitely many prime ideals $\mathfrak{m}_1, \dots, \mathfrak{m}_n$ each of which is also maximal, every element of $\bigcap \mathfrak{m}_i$ is nilpotent, and $\mathbb{Q}[\alpha]/\bigcap \mathfrak{m}_i$ is a product of fields

$$\mathbb{Q}[\alpha]/\bigcap \mathfrak{m}_i = K_1 \times \cdots \times K_n, \qquad K_i = \mathbb{Q}[\alpha]/\mathfrak{m}_i.$$

We first show that $\bigcap \mathfrak{m}_i = 0$, i.e., that $\mathbb{Q}[\alpha]$ has no nonzero nilpotents. Note that $\mathbb{Q}[\alpha]$ is stable under the action of $\dagger$. Let $a \neq 0 \in \mathbb{Q}[\alpha]$. Then $b \stackrel{\text{df}}{=} a^\dagger \cdot a \neq 0$, because $\mathrm{Tr}(a^\dagger \cdot a) > 0$. As $b^\dagger = b$, $\mathrm{Tr}(b^2) = \mathrm{Tr}(b^\dagger \cdot b) > 0$, and so $b^2 \neq 0$. Similarly, $b^4 \neq 0$, and so on, which implies b is not nilpotent, and so neither is a. Any automorphism τ of $\mathbb{Q}[\alpha]$ permutes the maximal ideals $\mathfrak{m}_i$; it therefore permutes the factors K_i, i.e., there is a permutation σ of $\{1, 2, \dots, n\}$ and isomorphisms $\tau_i : K_i \to K_{\sigma(i)}$ such that $\tau(a_1, \dots, a_n) = (b_1, \dots, b_n)$ with $b_{\sigma(i)} = \tau_i(a_i)$. In the case that $\tau = \dagger$, σ must be the trivial permutation, for otherwise $\mathrm{Tr}(a^\dagger \cdot a)$ would not always be positive (consider $(a_1, 0, 0, \dots)$ if $\sigma(1) \neq 1$). Hence $\dagger$ preserves the factors of $\mathbb{Q}[\alpha]$, and is a positive-definite involution on each of them. The involution $\dagger$ extends by linearity (equivalently by continuity) to a positive-definite involution of $\mathbb{Q}[\alpha] \otimes \mathbb{R}$. The above remarks also apply to $\mathbb{Q}[\alpha] \otimes \mathbb{R}$: it is a product of fields, and $\dagger$ preserves each factor and is a positive-definite involution on each of them. But now each factor is isomorphic to $\mathbb{R}$ or to $\mathbb{C}$. The field $\mathbb{R}$ has no nontrivial automorphisms at all, and so $\dagger$ must act on a real factor of $\mathbb{Q}[\alpha] \otimes \mathbb{R}$ as the identity map. The field $\mathbb{C}$ has only two automorphisms of finite order: the identity map and complex conjugation. The identity map is not positive-definite, and so $\dagger$ must act on a complex factor as complex conjugation. We have shown: for any homomorphism $\rho : \mathbb{Q}[\alpha] \to \mathbb{C}$, $\rho(\alpha^\dagger) = \overline{\rho(\alpha)}$. Thus, for any such homomorphism, $r = \rho(\alpha^\dagger \cdot \alpha) = |\rho(\alpha)|^2$, and so every root of the minimum polynomial of α in $\mathbb{Q}[\alpha]/\mathbb{Q}$ has absolute value $r^{1/2}$. Now (10.24) completes the proof. □

Source: 2008 notes, II Remark 1.4, physical page 84, printed page 78.

Remark 1.4. We have actually proved the following: $\mathbb{Q}[\pi]$ is a product of fields, stable under the involution $\dagger$; under every map $\tau : \mathbb{Q}[\pi] \to \mathbb{C}$, $\tau(\pi^\dagger) = \overline{\tau(\pi)}$, and $|\tau\pi| = q^{1/2}$.

Editorial note. The parent text's use of $V(\mathbb{F}_q)$ in the introduction, its $\mathbb{A}^m/\mathbb{A}^n$ notation and differential display in the initial proof, and the printed conclusion $\pi^* D' = qD$ in II1.2 are retained literally. The first proof gives the error-bound calculation for $m = 1$ as printed, while the statement retains the bound for every m. No additional general-m calculation or additional condition on r in II1.3 has been supplied. These are transcription notes, not mathematical repairs or a correctness assessment.

Separate supporting source: the corrected Storrs article

Version boundary. The following is from the fixed January 2, 2022 corrected author file AVs, not from the 2008 course notes. Physical pages and printed pages both read 35–36. The source retains

[1] In fact, it is easy to prove this directly. Let $f(X)$ be a monic polynomial generating the kernel of $\mathbb{Q}[X] \to \mathbb{Q}[\alpha]$. Then $\mathbb{Q}[X]/(f(X)) \simeq \mathbb{Q}[\alpha]$, and the maximal ideals of $\mathbb{Q}[\alpha]$ correspond to the distinct irreducible factors of $f(X)$.

its own numbering, notation l for the auxiliary prime, $\varphi_{\mathcal{L}}$, and the references to its own (15.4). The argument is an uncounted support for the particular trace/intersection formula omitted in the parent I14.3.

17. The Rosati Involution: necessary opening context

Source: corrected Storrs article, physical/printed page 35; opening paragraphs of Section 17.

Fix a polarization λ on A. As λ is an isogeny $A \to A^\vee$, it has an inverse in $\mathrm{Hom}^0(A^\vee, A) \stackrel{\mathrm{def}}{=} \mathrm{Hom}(A^\vee, A) \otimes \mathbb{Q}$. The *Rosati involution* on $\mathrm{End}^0(A)$ corresponding to λ is

$$\alpha \longmapsto \alpha^\dagger = \lambda^{-1} \circ \alpha^\vee \circ \lambda.$$

This has the following obvious properties:

$$(\alpha+\beta)^\dagger = \alpha^\dagger + \beta^\dagger, \qquad (\alpha\beta)^\dagger = \beta^\dagger\alpha^\dagger, \qquad a^\dagger = a \text{ for } a \in \mathbb{Q}.$$

For all $a, a' \in T_l A \otimes \mathbb{Q}$, $l \neq \mathrm{char}(k)$,

$$e_l^\lambda(\alpha a, a') = e_l(\alpha a, \lambda a') = e_l(a, \alpha^\vee \circ \lambda a') = e_l^\lambda(a, \alpha^\dagger a');$$

from which it follows that $\alpha^{\dagger\dagger} = \alpha$.

Source: corrected Storrs article, Theorem 17.3, physical/printed page 35.

Theorem 17.3. *The bilinear form*

$$(\alpha, \beta) \longmapsto \mathrm{Tr}(\alpha \circ \beta^\dagger) : \mathrm{End}^0(A) \times \mathrm{End}^0(A) \longrightarrow \mathbb{Q}$$

is positive definite. More precisely, if $\lambda = \varphi_{\mathcal{L}(D)}$, then

$$\mathrm{Tr}(\alpha \circ \alpha^\dagger) = \frac{2g}{(D^g)}(D^{g-1} \cdot \alpha^*(D)).$$

Source: corrected Storrs article, physical/printed page 36; paragraph continuing Theorem 17.3.

As D is ample and $\alpha^*(D)$ is effective, the intersection number $(D^{g-1} \cdot \alpha^*(D))$ is positive. Thus the second statement implies the first. Clearly it suffices to prove it with k algebraically closed. We first need a lemma.

Lemma 17.4. *Let A be an abelian variety over an algebraically closed field, and let $\mathbb{Z}_l(g) = \mathbb{Z}_l(1)^{\otimes g}$. Then there is a canonical generator ε of $\mathrm{Hom}(\bigwedge^{2g}(T_l A), \mathbb{Z}_l(g))$ with the following property: if $D_1, \dots, D_g$ are divisors on A and $e_i = e_l^{\mathcal{L}(D_i)} \in \mathrm{Hom}(\bigwedge^2 T_l A, \mathbb{Z}_l(1))$, then $e_1 \wedge \cdots \wedge e_g$ is the multiple $(D_1, D_2, \dots, D_g)\varepsilon$ of ε.*

Proof. See Mumford 1970, §20, Theorem 3, p. 190. (From the point of view of étale cohomology, ε corresponds to the canonical generator of $H^{2g}(A, \mathbb{Z}_l(g))$, which is equal to the cohomology class of any point on A. If c_i is the class of D_i in $H^2(A, \mathbb{Z}_l(1))$, then the compatibility of intersection products with cup products shows that $(D_1, \dots, D_g)\varepsilon = c_1 \cup \cdots \cup c_g$. Consequently, the lemma follows from (15.4). □

Source: corrected Storrs article, physical/printed page 36; complete calculation ending "which completes the proof".

Proof (of 17.3). From the lemma, we find that

$$e_l^\lambda \wedge \cdots \wedge e_l^\lambda = (D^g)\varepsilon,$$
$$e_l^\lambda \wedge \cdots \wedge e_l^\lambda \wedge e_l^{\alpha^*(\lambda)} = (D^{g-1} \cdot \alpha^*(D))\varepsilon.$$

It suffices therefore to show that, for some basis $a_1, \ldots, a_{2g}$ of $T_l A \otimes \mathbb{Q}$,

$$\frac{\langle a_1 \wedge \cdots \wedge a_{2g} \mid e_l^\lambda \wedge \cdots \wedge e_l^\lambda \wedge e_l^{\alpha^*(\lambda)} \rangle}{\langle a_1 \wedge \cdots \wedge a_{2g} \mid e_l^\lambda \wedge \cdots \wedge e_l^\lambda \rangle} = \frac{1}{2g} \operatorname{Tr}(\alpha \circ \alpha^\dagger)$$

(see 15.4). Choose the basis $a_1, a_2, \ldots, a_{2g}$ so that

$$\begin{aligned} e_l^\lambda(a_{2i-1}, a_{2i}) = 1 = -e_l^\lambda(a_{2i}, a_{2i-1}), &\qquad i = 1, 2, \ldots, g, \\ e_l^\lambda(a_i, a_j) = 0, &\qquad \text{otherwise.} \end{aligned}$$

Let $f_1, \ldots, f_{2g}$ be the dual basis; then for $j \neq j'$,

$$\langle a_i \wedge a_{i'} \mid f_j \wedge f_{j'} \rangle = \begin{vmatrix} f_j(a_i) & f_{j'}(a_i) \\ f_j(a_{i'}) & f_{j'}(a_{i'}) \end{vmatrix} = \begin{cases} 1 & \text{if } i = j,\ i' = j', \\ -1 & \text{if } i = j',\ i' = j, \\ 0 & \text{otherwise.} \end{cases}$$

Therefore $e_l^\lambda = \sum_{i=1}^g f_{2i-1} \wedge f_{2i}$, and so $e_l^\lambda \wedge \cdots \wedge e_l^\lambda = g!(f_1 \wedge \cdots \wedge f_{2g})$. Thus

$$\langle a_1 \wedge \cdots \wedge a_{2g} \mid e_l^\lambda \wedge \cdots \wedge e_l^\lambda \rangle = \langle a_1 \wedge \cdots \wedge a_{2g} \mid g!(f_1 \wedge \cdots \wedge f_{2g}) \rangle = g!.$$

Similarly,

$$\begin{aligned} &\langle a_1 \wedge \cdots \wedge a_{2g} \mid e_l^\lambda \wedge \cdots \wedge e_l^\lambda \wedge e_l^{\alpha^*(\mathcal{L})} \rangle \\ &\quad = (g-1)! \sum_{i=1}^{g} e_l^\lambda(\alpha a_{2i-1}, \alpha a_{2i}) \\ &\quad = \frac{(g-1)!}{2} \sum (e_l^\lambda(a_{2i-1}, \alpha^\dagger \alpha a_{2i}) + e_l^\lambda(\alpha^\dagger \alpha a_{2i-1}, a_{2i})) \\ &\quad = \frac{g!}{2g} \operatorname{Tr}(\alpha^\dagger \alpha), \end{aligned}$$

which completes the proof. □

Editorial note. The source's explicit algebraic-closure reduction and the auxiliary-prime condition $l \neq \operatorname{char}(k)$ are retained; there is no characteristic-zero or principal-polarization restriction added here. Lemma 17.4 retains both its Mumford reference and its printed cup-product explanation with the same article's (15.4). Those background interfaces are not re-proved. The source's unclosed explanatory parenthesis in that proof, its use of $e_l^{\alpha^*(\lambda)}$ followed later by $e_l^{\alpha^*(\mathcal{L})}$, and the final unindexed $\sum$ are kept as printed, without mathematical emendation.